LESSON PLAN

PISTON ENGINE (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Efficiency, BHP, IHP, FHP, SHP, knocking and detonation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Efficiency, BHP, IHP, FHP, SHP, knocking and detonation. Ignition system, basic principles, description and operations
5. Specific Objectives	15.1.3.1 Trainees will be able to explain various engine power

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine power definitions and efficiency calculations using IB data; Instructor supervision required for calculation verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the difference between Indicated Horsepower and Brake Horsepower?
(b) Define the term 'Friction Horsepower' in the context of piston engines.
(c) Why is thermal efficiency a critical parameter for engine performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Indicated Horsepower (IHP) • Definition of Brake Horsepower (BHP) • Calculation of Friction Horsepower (FHP) • Relationship between IHP, BHP, and FHP • Concept of Mechanical Efficiency	L & Ds	PPT, IB	15 min
(b)	• Definition of Shaft Horsepower (SHP) • Calculation of Brake Thermal Efficiency • Identification of Knocking phenomena • Distinction between Knocking and Detonation • Causes and effects of Detonation	L & Ds	PPT, PDF	20 min
(c)	• Impact of detonation on engine components • Methods to prevent knocking and detonation • Role of octane rating in fuel selection • Effect of supercharging on detonation tendency • Summary of power definitions and efficiency metrics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how Friction Horsepower is derived from Indicated and Brake Horsepower.
(b) Describe the primary difference between knocking and detonation in a piston engine.
(c) List three operational factors that increase the tendency for detonation.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operation of a simple float type carburetor with a modified carburetor, focusing on how fuel-air mixture control affects engine efficiency and detonation risk, as covered in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Description & operation of simple float type carburetor, modified carburetor, limitation Types of Engine starters, & trouble shooting on piston engine	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min